

PROBABILITY, (CONDITIONAL PROBABILITY AND VENN

Name: _____

Date: _____

LO: I can use Venn diagrams to find conditional probabilities.

Conditional probability

Conditional probability is the probability of an event occurring **given that** another event has already occurred. Written $P(A|B)$ meaning 'probability of A given B'.

A **Venn diagram** shows overlapping sets. The **intersection** shows elements in both sets.

Reading Venn diagrams

●	$P(A B) = P(A \text{ and } B) \div P(B)$	conditional formula
●	Intersection = elements in BOTH sets	overlap region
●	$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$	addition rule
●	$P(A B) = P(A) \times P(B)$	that is for independent events only

Key idea

$P(A|B)$ means restrict to only those outcomes in B, then find how many are also in A.

$$P(A|B) = \frac{n(A \cap B)}{n(B)}$$

Restrict the sample space to B

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - from a Venn diagram

In a class: 12 play football, 8 play tennis, 5 play both, 30 total. Find $P(\text{football} | \text{tennis})$.

Those who play tennis = 8
Of those, 5 also play football
 $P(\text{football} | \text{tennis}) = \frac{5}{8}$

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Given they play tennis (8 people), how many also play football (5).

Example 2 - $P(A \text{ or } B)$

Using the same data, find $P(\text{football or tennis})$.

$$\begin{aligned} P(F \text{ or } T) &= P(F) + P(T) - P(F \text{ and } T) \\ &= \frac{12}{30} + \frac{8}{30} - \frac{5}{30} \\ &= \frac{15}{30} = \frac{1}{2} \end{aligned}$$

$$P(\text{football or tennis}) = \frac{1}{2}$$

Subtract the intersection to avoid counting it twice.

Example 3 - neither event

Find $P(\text{neither football nor tennis})$.

$$\begin{aligned} \text{In at least one sport: } &12 + 8 - 5 = 15 \\ \text{Neither: } &30 - 15 = 15 \\ P(\text{neither}) &= \frac{15}{30} = \frac{1}{2} \end{aligned}$$

$$P(\text{neither}) = \frac{1}{2}$$

Subtract those in at least one set from the total.

Example 4 - given information

$P(A) = 0.6$, $P(B) = 0.4$, $P(A \text{ and } B) = 0.2$. Find $P(B|A)$.

$$\begin{aligned} P(B|A) &= P(A \text{ and } B) \div P(A) \\ &= 0.2 \div 0.6 \\ &= \frac{1}{3} \end{aligned}$$

$$P(B|A) = \frac{1}{3}$$

Divide the intersection probability by the condition probability.

YOUR TURN – PRACTICE (deliberate practice)

1. Reading Venn diagrams

A Venn diagram shows: A only = 7, B only = 5, A and B = 3, neither = 5. Total = 20.

- a) $P(A)$
- b) $P(B)$
- c) $P(A \text{ and } B)$
- d) $P(A \text{ or } B)$

2. Conditional probability

Using the same data: A only = 7, B only = 5, both = 3, neither = 5.

- a) $P(A | B)$
- b) $P(B | A)$
- c) $P(A | \text{not } B)$
- d) $P(\text{neither} | \text{not } A)$

3. From probabilities

$P(A) = 0.5$, $P(B) = 0.3$, $P(A \text{ and } B) = 0.1$. Find:

- a) $P(A \text{ or } B)$
- b) $P(A | B)$
- c) $P(B | A)$
- d) $P(\text{neither } A \text{ nor } B)$

4. Two-way tables to Venn diagrams

60 students: 25 study French, 20 study Spanish, 10 study both.

- a) $P(\text{French only})$
- b) $P(\text{Spanish} | \text{French})$
- c) $P(\text{at least one language})$
- d) $P(\text{French} | \text{Spanish})$

5. Mixed conditional problems

Solve each conditional probability problem.

- a) $P(A) = 0.7$, $P(A \text{ and } B) = 0.35$. Find $P(B | A)$.
- b) $P(B) = 0.4$, $P(A | B) = 0.5$. Find $P(A \text{ and } B)$.
- c) 100 people: 40 like tea, 30 like coffee, 15 like both. $P(\text{tea} | \text{coffee}) = 0.4$. Find $P(A \text{ and } B)$.
- d) $P(A \text{ or } B) = 0.8$, $P(A) = 0.5$, $P(B) = 0.4$. Find $P(A \text{ and } B)$.
- e) $P(\text{pass} | \text{revised}) = 0.9$, $P(\text{revised}) = 0.6$. Find $P(\text{pass and revised})$.

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) $P(A | B)$ means the probability of A given B has occurred ☐

Correct answer: _____

Reason: _____

b) $P(A \text{ and } B) = P(A) + P(B)$ ☐

Correct answer: _____

Reason: _____

c) $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$ ☐

Correct answer: _____

Reason: _____

d) $P(A | B) = P(B | A)$ always ☐

Correct answer: _____

Reason: _____

e) In a Venn diagram, the intersection shows elements in both sets ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) In a school, **70%** of students have a phone, **50%** have a tablet, and **30%** have both. A student is chosen at random. Given they have a phone, what is the probability they also have a tablet?

Expression: _____

Simplified: _____

b) A medical test is **95% accurate** ($P(\text{positive} | \text{disease}) = 0.95$). The disease affects **1%** of the population. If a person tests positive, what is $P(\text{disease} | \text{positive})$? (Hint: consider false positives with $P(\text{positive} | \text{no disease}) = 0.05$.)

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

